

Action-Conditioned Predictive Diagnostics for JEPA World Models

Author names to be supplied for arXiv v1

July 2026

Abstract

Visual changes that leave the underlying control state unchanged can still alter the futures predicted by a world model. Encoder-level invariance alone does not show whether learned dynamics amplify these changes or whether they affect planning. We study this problem with Action-Conditioned Predictive Consistency (ACPC), which rolls nominal and visually perturbed histories forward under the same action sequence and compares their predicted representations. ACPC gives exact inequalities for changes in future prediction error and squared goal cost, together with a different-state guard against trivial collapse. Across four control tasks, recorded-action eight-step ACPC reduces the error of predicting changes in eight-step latent error by $51.3 \pm 3.5\%$ relative to the strongest same-horizon control using zeroed or shuffled actions. Candidate-conditioned ACPC also improves prediction of planner cost changes and decision regret. Thresholds selected on source tasks reach 0.84–0.86 balanced accuracy on the remaining tasks, and the resulting rule preserves useful checkpoint ordering under blur and resize without shift-specific adjustment. The same equations and evaluation procedure apply to PLDM, with thresholds selected separately for that model family.

Keywords: visual world models; predictive stability; action-conditioned diagnostics; calibration transfer; visual corruption.

Code and data release: <https://github.com/Anguo-star/lewm-acpc-diagnostics>

1 Introduction

World models support control by predicting the outcomes of candidate actions [11, 12]. Joint-embedding predictive architectures (JEPAs) replace pixel reconstruction with prediction in representation space [16, 1, 3]. This can reduce pressure to model task-irrelevant visual detail [28, 17], which is attractive when texture, lighting, or image quality changes without changing the control state.

Robust control, however, does not call for indiscriminate invariance. Background texture may be a nuisance, but a contact pose, doorway state, or object–goal relation must remain distinguishable when it changes the useful action. A planner also consumes predictions rather than an encoder output alone: a small encoding discrepancy can grow through learned dynamics, while a large one can contract before the cost readout. The relevant question is therefore whether the discrepancy introduced by a same-state visual perturbation remains small after the same action sequence.

Consistency must also be selective. Mapping every observation to the same point would make nominal and perturbed inputs agree perfectly while destroying information needed for action choice. A useful diagnostic must combine same-state stability with evidence that genuinely different states remain separated [8, 33, 24, 10]. To inform planning, it should also indicate when predictive disagreement can change candidate costs or decisions.

We study these requirements with Action-Conditioned Predictive Consistency (ACPC). This is a diagnostic study of frozen checkpoints, not a new robust-training method. ACPC rolls a nominal history and a visually perturbed version of the same state through the predictor under a shared action sequence, measuring projected latent disagreement without a realized future or task outcome. For each pair, exact inequalities bound changes in future prediction error and, for squared-distance goals, candidate cost. A different-state margin guard rejects trivial low-radius collapse. The same predictive interface can be used across inspectable models, although numerical thresholds may remain model-family specific.

We evaluate ACPC on LeWM [19] under Gaussian observation noise across TwoRoom, PushT, Reacher, and Cube, and apply the same analysis to a complete PLDM sweep. When evaluated at observation noise $\sigma = 0.08$,

LeWM checkpoints trained without noise augmentation lose up to 74.56 ± 5.55 percentage points in planning success rate. We test whether recorded-action eight-step ACPC predicts the change in future error better than one-step ACPC and same-horizon controls with zeroed or shuffled actions. We then examine its relation to CEM candidate costs and decisions. Finally, we select thresholds on every source-task subset, evaluate them on the remaining tasks, and test the fixed Gaussian-noise rule under blur and resize.

The contributions are:

1. **A target-free diagnostic of predictive sensitivity.** We define ACPC by rolling nominal and visually perturbed histories forward under the same actions. Exact bounds relate its disagreement score to future-error and goal-cost changes, while a different-state margin rules out trivial collapse.
2. **Controlled evidence for prediction and planning relevance.** Across four tasks and training runs, recorded-action eight-step ACPC predicts visual-perturbation-induced error changes more accurately than one-step ACPC and same-horizon controls with zeroed or shuffled actions. Candidate-conditioned ACPC also adds information about CEM cost changes and decision regret.
3. **Evaluation across tasks, visual shifts, and model families.** We evaluate every source/remaining-task partition, apply one Gaussian-noise calibration to blur and resize without shift-specific adjustment, and run the same analysis on PLDM. The procedure is shared, while numerical thresholds are selected separately for each model family.

2 Related Work

Prior work relevant to ACPC falls into three groups: predictive world models, methods for visual robustness, and diagnostics of learned dynamics.

Predictive world models. World models learn dynamics for predicting the consequences of candidate actions [11, 12]. JEPAs predict representations rather than pixels, as developed for images and video by I-JEPA and V-JEPA [16, 1, 3, 2]. LeWM builds an end-to-end JEPA world model stabilized by SIGReg [19, 7]; PLDM provides a different joint-embedding dynamics architecture, implemented here through `stable-worldmodel` [25, 26, 18]. Analyses of latent prediction explain how it can suppress nuisance or noisy features [28, 17, 14], while Seq-JEPA studies the balance between invariant and equivariant prediction [9]. Our focus is a frozen model: we measure how visual differences propagate through its predictor under actions.

Visual robustness for control. DrQ, DrQ-v2, and SODA improve visual control through training-time augmentation [15, 32, 13]. ViGMO and VIBR instead learn latent or value-function invariance to visual changes [20, 6], while ReOI operates at test time to reduce distractor sensitivity in visual model-predictive control [5]. A complementary line of work uses bisimulation or value-aware objectives to preserve state distinctions that change downstream control decisions [8, 33, 24, 10, 29, 27]. ACPC does not modify training or repair observations. It evaluates a frozen checkpoint by combining same-state rollout agreement with a different-state separation guard.

Diagnostics of learned dynamics. Recent diagnostics test whether learned dynamics preserve action and temporal structure through inverse models, latent action decoding, cycle consistency, or compatibility with generated futures [4, 34, 23, 21, 30]. MWM incorporates action-conditioned consistency into training [31], whereas iKCE identifies kinematic failures in long-horizon rollouts [22]. These studies motivate examining the predictor rather than the encoder alone. ACPC complements them by comparing two visual views of the same state under shared actions and relating their rollout disagreement to the change in future prediction error.

3 Action-Conditioned Predictive Consistency

A same-state visual corruption first creates an upstream latent displacement. For control, that displacement is only a proxy: the planner acts on predicted futures and costs after the action-conditioned predictor has had a chance to amplify, rotate, or contract the shift. At the same time, useful control requires selectivity. A background perturbation should not move the predicted future much, but an object pose, doorway state, contact state, or goal relation that changes the useful action must remain distinguishable.

ACPC is built around this radius–margin view. The radius side asks whether two visual views of the same underlying state remain inside a small rollout tube under the same actions. The margin side asks whether task-different pairs stay outside that tube. The pairwise ACPC radius gives the prediction-error and planner-cost statements below; ATR summarizes this radius over logged anchors; SMPR is the empirical margin guard against low-radius collapse.

3.1 From visual perturbations to rollout tubes

Let h be a nominal observation–action history and let $\tilde{h}$ be a visually perturbed view of the same underlying control state. A frozen encoder maps them to $z = E_\theta(h)$ and $\tilde{z} = E_\theta(\tilde{h})$. The predictor F_θ rolls both representations forward under one shared action sequence $\mathbf{a} = (a_0, \dots, a_{H-1})$. A projection Π selects the representation component used by the diagnostic, and nonnegative weights α_k with $\sum_{k=1}^H \alpha_k = 1$ combine the H future steps. Define

$$\bar{G}_{\mathbf{a}}(z) = [\sqrt{\alpha_1}\Pi(F_\theta^1(z, a_0)); \dots; \sqrt{\alpha_H}\Pi(F_\theta^H(z, \mathbf{a}))], \quad \text{ACPC}_H(h, \tilde{h}, \mathbf{a}) = \|\bar{G}_{\mathbf{a}}(z) - \bar{G}_{\mathbf{a}}(\tilde{z})\|_2. \quad (1)$$

The map $\bar{G}_{\mathbf{a}}$ is the weighted predicted rollout. ACPC is the distance between the clean and perturbed rollouts after the same action intervention; it is the radius of the same-state rollout tube. This differs from encoder shift, which is measured before actions are applied and therefore cannot tell whether the predictor will amplify or contract the visual displacement. Here H counts future prediction steps, not history frames. *Action matched* means that both branches receive the same recorded or candidate actions; it isolates visual sensitivity under a fixed intervention and does not by itself certify action faithfulness. The unnormalized ACPC in Equation (1) is the object used by the propositions. ATR below is the normalized checkpoint-level summary used for screening.

3.2 Common-future error drift

The first proposition is a bookkeeping guarantee. Once two branches are compared with the same action-matched future, their difference in prediction error cannot exceed their rollout disagreement. The future $Y_{\mathbf{a}}^H$ is not produced by either branch. It is an external target used only for evaluation, such as the projected latent of the actually observed future under the recorded action sequence, stacked and weighted like $\bar{G}_{\mathbf{a}}$. If the two branches are compared with different futures, the reverse-triangle argument no longer describes visual sensitivity.

Proposition 1 (Common-future error-drift bound). *Let $Y_{\mathbf{a}}^H$ be any projected external future generated from the same state and action sequence as both prediction branches. With*

$$e_h = \|\bar{G}_{\mathbf{a}}(E_\theta(h)) - Y_{\mathbf{a}}^H\|_2, \quad e_{\tilde{h}} = \|\bar{G}_{\mathbf{a}}(E_\theta(\tilde{h})) - Y_{\mathbf{a}}^H\|_2,$$

the following holds sample by sample:

$$|e_{\tilde{h}} - e_h| \leq \text{ACPC}_H(h, \tilde{h}, \mathbf{a}). \quad (2)$$

The adverse increase $(e_{\tilde{h}} - e_h)_+$ obeys the same bound. The right-hand side can be computed without a future target at diagnostic time.

Proof. This is the reverse triangle inequality in the weighted-stacked projected-rollout space. The common action sequence is required because both errors must share the same target. $\square$

The proposition bounds the difference between the two errors, not their absolute magnitude. A small ACPC value therefore does not prove that either prediction is accurate. It says only that, for a common action-matched future, the visual perturbation cannot change the prediction error by more than the paired-rollout radius. This is why the future-drift experiment predicts the absolute change in eight-step latent error rather than claiming that low ACPC means low prediction error.

3.3 Planner flips as radius–margin events

A planner flip is also a radius–margin event. A visual perturbation can change a fixed-pool CEM decision only if it moves candidate costs enough to cross a clean candidate gap. For LeWM’s implemented goal cost, only the final predicted representation at the candidate horizon enters the squared distance to the goal. We

therefore write $x_j = \Pi(F_\theta^H(E_\theta(h), \mathbf{a}^j))$ and $\tilde{x}_j = \Pi(F_\theta^H(E_\theta(\tilde{h}), \mathbf{a}^j))$ for the final clean and perturbed endpoints of candidate j in a fixed pool $\{\mathbf{a}^j\}_{j=1}^K$. Their displacement is $r_j = \|x_j - \tilde{x}_j\|_2$, the final-step component of ACPC. When $\alpha_H > 0$, $r_j \leq \text{ACPC}_H / \sqrt{\alpha_H}$, but the planner audit records r_j directly.

Proposition 2 (Squared goal-cost drift and fixed-pool certificates). *LeWM scores candidate j against a fixed goal embedding g by $C_j = \|x_j - g\|_2^2$ and $\tilde{C}_j = \|\tilde{x}_j - g\|_2^2$. Define*

$$b_j = r_j(\|x_j - g\|_2 + \|\tilde{x}_j - g\|_2). \quad (3)$$

Candidate cost. *Every candidate satisfies $|\tilde{C}_j - C_j| \leq b_j$.*

Decision regret. *Let w and $\tilde{w}$ be the nominal and perturbed winners of the same pool. The clean-history decision regret satisfies*

$$0 \leq C_{\tilde{w}} - C_w \leq b_{\tilde{w}} + b_w. \quad (4)$$

Winner stability. *If w is unique and*

$$\min_{j \neq w} (C_j - C_w - b_j - b_w) > 0, \quad (5)$$

then the perturbed branch has the same winner.

Elite-set stability. *If $\mathcal{E}$ is the nominal top- k elite set and*

$$\min_{i \in \mathcal{E}, j \notin \mathcal{E}} (C_j - C_i - b_j - b_i) > 0, \quad (6)$$

the perturbed branch has exactly the same elite set.

The term b_j is the most that candidate j 's squared goal cost can move from the observed endpoint displacement and goal distances alone. The regret bound says that choosing the perturbed winner on the clean branch cannot cost more than the movement allowed for the clean and perturbed winners. The winner and elite certificates are margin tests: after subtracting the allowed cost movement from each clean gap, a positive remainder rules out that fixed-pool flip. Unlike a global Lipschitz argument, Equation (3) uses the actual paired endpoints. It is worst-case tight from these quantities alone; proofs and equality cases are in Appendix A. The certificate is not a compute-saving rule because it requires evaluating the candidate under both matched histories.

The cross-entropy method (CEM) repeatedly fits its next proposal to the current elite set. With common random numbers, two identical proposals generate the same ordered candidate pool, so a certified common elite set also produces the same next proposal.

Corollary 1 (Conditional CEM stability). *Consider nominal and perturbed CEM runs initialized by the same proposal and sampled with common random numbers. While their ordered candidate pools are aligned, if Equation (6) holds at the current iteration, both runs fit the same next proposal. If it holds at every iteration, the pools, proposals, and final action remain aligned by induction.*

The corollary is conditional. It applies only while the two runs keep aligned candidate pools and stops at the first iteration where the elite certificate fails. It is a shared-pool model-cost statement, not a closed-loop return guarantee. Appendix G specifies the fixed-pool and adaptive-CEM evaluation contract.

3.4 Selective consistency and semantic margins

Small radius alone is insufficient because a constant representation would make every same-state pair agree while erasing the distinctions a planner needs. Selective consistency therefore couples a same-state tube with a different-state margin. Let $\mathcal{P}_{\text{same}}$ contain nominal/probed histories of the same state and let $\mathcal{P}_{\text{diff}}$ contain proxy pairs that cross a task-relevant state distinction, such as different positions, contact configurations, or goal relations. Let d_θ denote their normalized rollout distance.

Definition 1 (Selective predictive consistency). For radius r , margin $\gamma > 0$, and pass probabilities $p_{\text{same}}, p_{\text{diff}}$, a representation is selectively consistent when

$$\Pr_{\mathcal{P}_{\text{same}}} [d_\theta \leq r] \geq p_{\text{same}}, \quad \Pr_{\mathcal{P}_{\text{diff}}} [d_\theta \geq r + \gamma] \geq p_{\text{diff}}. \quad (7)$$

The first inequality accepts same-state visual invariance only inside radius r . The second inequality requires different-state proxy pairs to remain at least γ outside that tube. Selective Margin Pass Rate (SMPR) is the empirical pass rate of the second inequality when r is the same-state q90 tube. SMPR is designed to detect gross collapse; it does not prove semantic validity, action faithfulness, or preservation of every task-relevant distinction. Its role is to keep the radius diagnostic from rewarding a model that becomes stable by making everything look the same.

3.5 From pairwise quantities to checkpoint-level scores

The propositions above operate on one paired history or one fixed candidate pool. To compare checkpoints, we convert pairwise radii into checkpoint-level scores by normalizing each pair with a clean transition scale and summarizing the upper tail across logged anchors. For anchor i , let s_i be the median norm of its clean observed transitions, including the history-to-future boundary:

$$s_i = Q_{0.50}(\{\|u_{i,k}^{\text{obs}} - u_{i,k-1}^{\text{obs}}\|_2\}_{k=1}^H). \quad (8)$$

For probe draw m , define the normalized radius $R^{(i,m)}$ and the checkpoint-level *Action-Conditioned Trajectory Radius* (ATR):

$$R^{(i,m)} = \frac{\|\bar{G}_{\mathbf{a}_i}(E_\theta(h_i)) - \bar{G}_{\mathbf{a}_i}(E_\theta(\tilde{h}_i^{(m)}))\|_2}{s_i + \varepsilon}, \quad \text{ATR}_q = Q_q\left(\left\{\frac{1}{M} \sum_{m=1}^M R^{(i,m)}\right\}_{i=1}^n\right). \quad (9)$$

The numerator is the unnormalized ACPC radius used by the pairwise bounds. The scale s_i is an anchor-specific yardstick fixed by the clean logged transition, not an outcome label. ATR is the high-quantile same-state tube radius across logged anchors. We use q90 because a small number of high-radius anchors can still create fragile planner histories; q90 is a reporting summary, not a candidate-distribution probability and not a calibrated tail bound.

The construction requires paired views, an inspectable encoder–predictor path, and one common projection and normalization within a comparison. It does not assume LeWM-specific layers, losses, or representation geometry. For model family $\mathcal{F}$ and stressor v , thresholds t_R and t_M combine the radius side and the margin-guard side into one score:

$$S_{\mathcal{F},v}(\theta) = \min\left\{\frac{t_R - \text{ATR}_v(\theta)}{|t_R| + \varepsilon}, \frac{\text{SMPR}_v(\theta) - t_M}{|t_M| + \varepsilon}\right\}. \quad (10)$$

The minimum makes the score selective: one failing side makes the whole score negative. Thus $S \geq 0$ means that the checkpoint passes both calibrated conditions, $\text{ATR} \leq t_R$ and $\text{SMPR} \geq t_M$. Computing S for a fixed checkpoint requires no task outcome, although success rates are used when thresholds are selected on source tasks. Thresholds are learned separately for each model family and visual-shift calibration. Given a reference checkpoint θ_0 , the paired comparison is

$$\Delta S_{\mathcal{F},v}(\theta_1; \theta_0) = S_{\mathcal{F},v}(\theta_1) - S_{\mathcal{F},v}(\theta_0) > 0 \quad (11)$$

Both checkpoints use the same scoring map, so $\Delta S > 0$ asks only whether θ_1 improves on θ_0 under that map. The rule requires the reference and does not imply that either checkpoint is absolutely robust.

Definition 2 (Calibrated diagnostic region). For a fixed model family and visual-shift calibration, the diagnostic region contains checkpoints satisfying $\text{ATR} \leq t_R$ and $\text{SMPR} \geq t_M$. Its meaning is evaluated on tasks excluded from threshold selection; raw thresholds are not assumed to transfer across model architectures.

Table 1 summarizes the objects and where they enter the experiments. The prediction experiment tests the common-future error-drift proposition by asking whether recorded-action eight-step ACPC predicts the change in eight-step latent error better than one-step ACPC or same-horizon action-destroyed controls. The planner experiment tests the fixed-pool and adaptive-CEM quantities. The cross-task, cross-stressor, and PLDM analyses use the checkpoint-level ATR+SMPR score to test calibrated threshold transfer. These are separate claims: prediction-error sensitivity, planner relevance, and calibrated checkpoint screening are not interchangeable.

Table 1: Reader guide to the ACPC objects.

Object	Intuition	Used in
ACPC	Same-state rollout-tube radius under a shared action sequence.	Prediction and planner diagnostics
Common-future drift	How much the visual probe changes error to one action-matched future.	Future-drift experiment
b_j	Maximum squared-cost movement allowed for candidate j from its paired endpoints.	Fixed-pool planner certificates
ATR	q90 normalized same-state tube radius over logged anchors.	Checkpoint screening
SMPR	Fraction of different-state proxies that remain outside the same-state tube.	Collapse guard
S	Worse side of the calibrated radius and margin passes.	Cross-task threshold transfer
ΔS	Change in S relative to a reference checkpoint under the same scoring map.	Cross-stressor checkpoint ordering

4 Experiments

4.1 Evaluation setup

Models, tasks, and planning evaluation. LeWM [19] is the primary model family; PLDM [25, 26] provides a second joint-embedding dynamics architecture. We evaluate TwoRoom navigation, PushT planar manipulation, Reacher arm control, and OGBench-Cube (Cube) 3D manipulation. We follow the standard LeWM latent-space MPC evaluation: CEM optimizes candidate action sequences with the frozen world model, and performance is the percentage of evaluation episodes that reach the task goal, reported as *planning success rate*. The main evaluation adds Gaussian noise $\sigma = 0.08$ to observations while keeping the goal image clean. Additional evaluations use Gaussian blur ($k = 15$) and resize (scale 0.25).

For each model family, the Gaussian-augmentation sweep contains a checkpoint trained without noise augmentation and checkpoints trained with full-sequence noise $\sigma_{\max} \in \{0.01, \dots, 0.08\}$. Each LeWM task-augmentation setting is trained with seeds 3072/3073/3074 and evaluated with seeds 42/43/44 using 100 episodes per evaluation seed. Evaluation seeds are conditional measurement replicates; the independently trained model is the replication unit. PLDM uses one independently trained model family over the same four tasks and nine augmentation levels. Its 36 checkpoints use the same evaluation seeds and episode count.

Diagnostic measurements. ATR uses 100 logged anchors, five perturbation draws, eight predicted steps, uniform horizon weights, within-anchor draw averaging, and a q90 summary across anchors. SMPR uses the same-state q90 tube, the closest-35% different-state proxy neighborhood, and margin 0.10. Appendix B gives the complete normalization and threshold protocol.

Prediction and CEM analyses. To test prediction-error changes, we use 16 trajectory groups per task, four perturbation severities, and two perturbation draws. The target is the absolute change in eight-step latent prediction error between nominal and perturbed histories. A ridge model is fitted on 15 groups and evaluated on the remaining group, rotating through all 16. Every regression uses perturbation severity, encoder shift, action magnitude, and nominal dynamics. The regressions differ only in the added ACPC feature: one-step ACPC, the strongest eight-step control using zeroed or shuffled actions, or eight-step ACPC using the recorded actions.

For the CEM analysis, checkpoints trained without noise augmentation and with $\sigma_{\max} = 0.08$ are evaluated on the same candidate pools. The reduced-budget analysis uses 100 histories per task and training run, $K = 64$, top-eight elites, and eight CEM steps. The full-budget analysis uses the same 16 histories per task with the deployed $K = 300$, top-30, 30-step configuration. Within each training run, ridge models are fitted on three tasks and evaluated on the fourth. We test whether candidate-conditioned five-step ACPC improves prediction of the maximum candidate-cost change, first-action RMS, and clean-history decision regret beyond perturbation severity, training condition, one-step ACPC, and the nominal top-1 margin.

Threshold evaluation. A Gaussian-sweep checkpoint meets the success-rate criterion when it attains at least 80% of the improvement from the checkpoint trained without augmentation to the best checkpoint at evaluation noise $\sigma = 0.08$, while losing no more than five percentage points on clean observations. We select (t_R, t_M) on every nonempty proper subset of the four tasks and apply the thresholds unchanged to the remaining tasks. This gives 14 directional source/evaluation partitions. Metrics first average training runs within each evaluation

task and then weight tasks equally; checkpoint rows are not treated as independent replicates. Success rates are used to select and evaluate thresholds, but they are not inputs to ACPC, ATR, or SMPR.

For blur and resize, each task uses the Gaussian thresholds selected on the other three tasks. We compare all 24 pairs of checkpoints trained without augmentation and with $\sigma_{\max} = 0.08$. A pair is positive when success rate under blur or resize improves by at least five percentage points and clean success rate decreases by no more than five points. No blur- or resize-specific adjustment is made, and the transfer analysis does not rank encoder shift or one-step ACPC.

4.2 Planning performance under observation noise

At evaluation noise $\sigma = 0.08$, LeWM checkpoints trained without noise augmentation lose 25.89 ± 2.38 , 74.56 ± 5.55 , 41.00 ± 1.44 , and 22.11 ± 2.78 percentage points in success rate on TwoRoom, PushT, Reacher, and Cube, respectively. Gaussian augmentation improves performance over a range of training levels whose location differs by task. Figure 1 compares success rate with ATR and SMPR across the complete sweep.

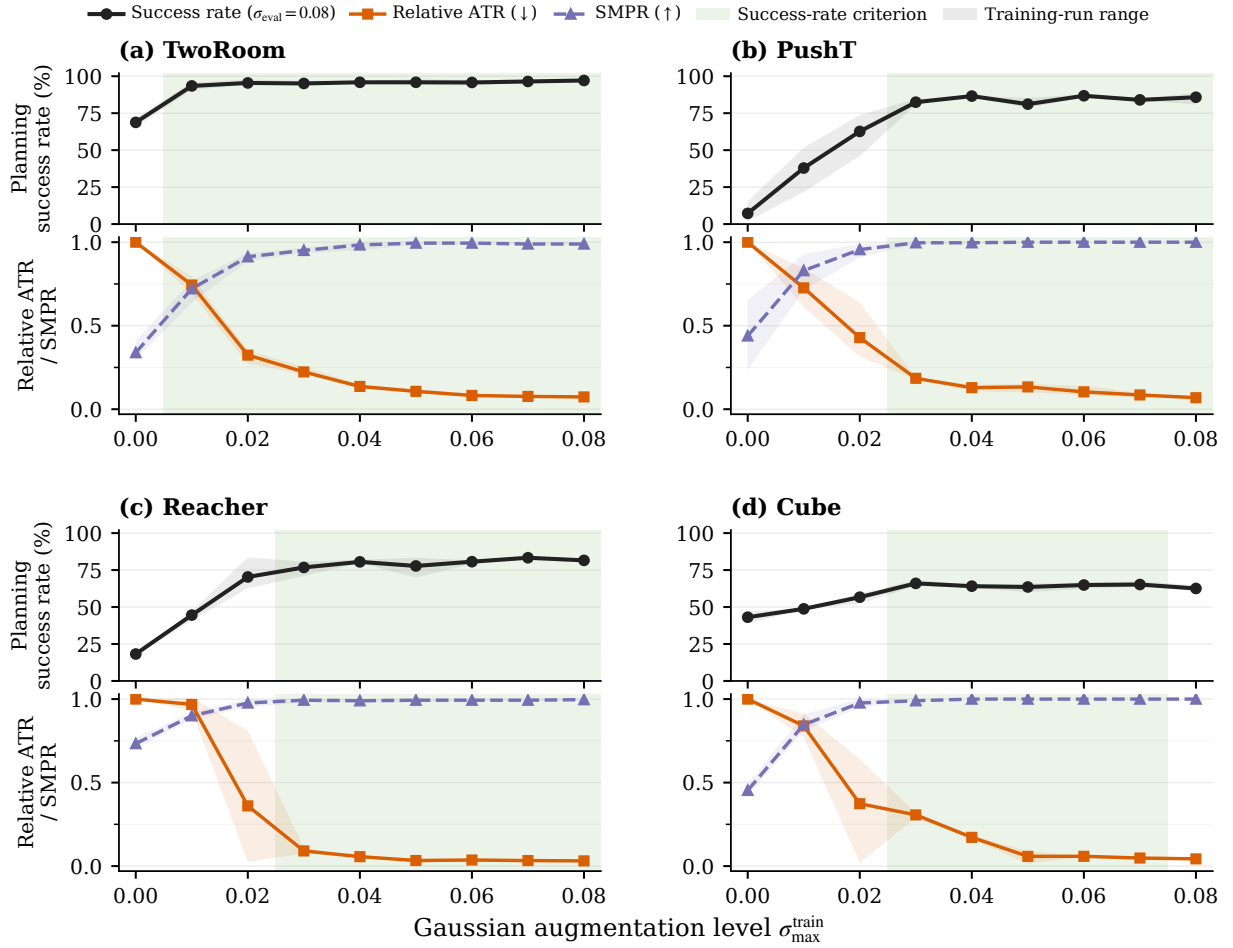


Figure 1: Planning success rate and diagnostic measurements across the Gaussian-augmentation sweep. Success rates use evaluation noise $\sigma = 0.08$; ATR is relative to the checkpoint trained without augmentation (value 1), and SMPR remains on its original scale. Lines show across-run means, and shaded ranges span the run means. Green bands mark levels that satisfy the success-rate criterion in a majority of runs.

ATR decreases and SMPR increases over the levels that meet the success-rate criterion on every task. This association motivates the combined diagnostic, but it does not by itself show that ACPC uses action information or predicts a planning-relevant quantity. The next two experiments test those questions.

4.3 Predicting error changes under visual perturbations

We first test the quantity bounded by Proposition 1: the absolute change in eight-step latent prediction error between a nominal history and a visually perturbed history, both compared with the same observed future. Every regression predicts this target from the same perturbation severity, encoder shift, action magnitude, and nominal-dynamics features. The only difference is the ACPC feature: one-step ACPC; the better of the two eight-step controls with zeroed or shuffled actions; or eight-step ACPC using the recorded actions. The same-horizon comparison isolates action information from rollout length.

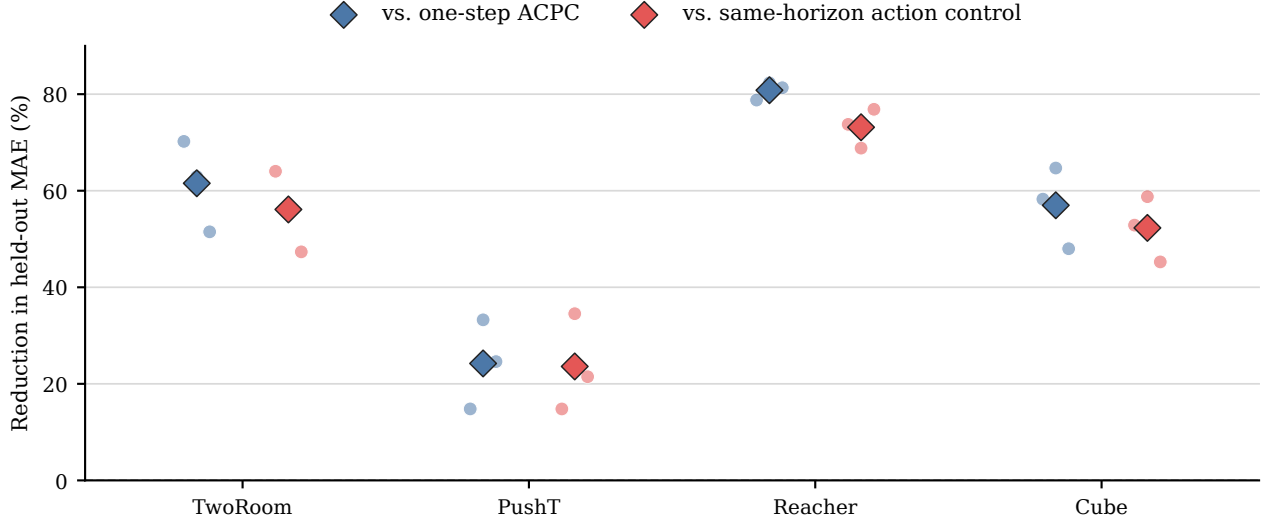


Figure 2: Reduction in held-out error when recorded-action eight-step ACPC replaces each comparator. Small circles show individual training runs and diamonds show the mean for each task. Both comparisons predict the same absolute change in eight-step latent prediction error. The same-horizon control uses whichever of zeroed or shuffled actions performs better.

Recorded-action eight-step ACPC reduces held-out MAE by $55.9 \pm 4.7\%$ relative to one-step ACPC and by $51.3 \pm 3.5\%$ relative to the strongest same-horizon control (equal-task mean $\pm$ sample SD across runs). All 12 task-run comparisons improve. These results concern the ability to predict a change in eight-step error; they do not compare the absolute accuracy of one-step and eight-step predictions. Appendix C reports the task-level MAEs and selected controls.

4.4 ACPC and CEM decisions

We next test whether ACPC is informative about the CEM quantities in Proposition 2. Across 9,600 reduced-budget records, the candidate-cost bound has no violations, and the winner and elite-set certificates never assert stability when the corresponding set changes. Identity probes give zero five-step ACPC and zero first-action RMS. Whenever the elite-set certificate remains valid, the nominal and perturbed CEM updates also remain aligned, as required by Corollary 1.

Table 2: Added predictive value of candidate-conditioned five-step ACPC on 9,600 reduced-budget history records. Within each training run, ridge models are fitted on three tasks and evaluated on the fourth. Both models use perturbation severity, training condition, one-step ACPC, and the nominal top-1 margin; the expanded model also uses five-step ACPC. Entries are reductions in held-out MAE, averaged equally across evaluation tasks.

Response	Mean $\pm$ SD	Run range	Cells improved
Maximum fixed-pool cost change	$7.9 \pm 1.4\%$	6.6–9.5%	8/12
Adaptive CEM decision regret	$15.2 \pm 2.0\%$	12.9–16.9%	12/12
Adaptive first-action RMS	$1.1 \pm 0.5\%$	0.8–1.7%	10/12

Adding candidate-conditioned five-step ACPC reduces held-out MAE for maximum fixed-pool cost change by $7.9 \pm 1.4\%$; the reduction is positive on three of the four tasks. For adaptive CEM, it reduces held-out MAE for positive decision regret by $15.2 \pm 2.0\%$, with an improvement in every task-run cell. The $1.1 \pm 0.5\%$ reduction

for first-action RMS is too small to support an incremental claim. In all three analyses, the rank association is higher for five-step than for one-step ACPC in every task-run cell.

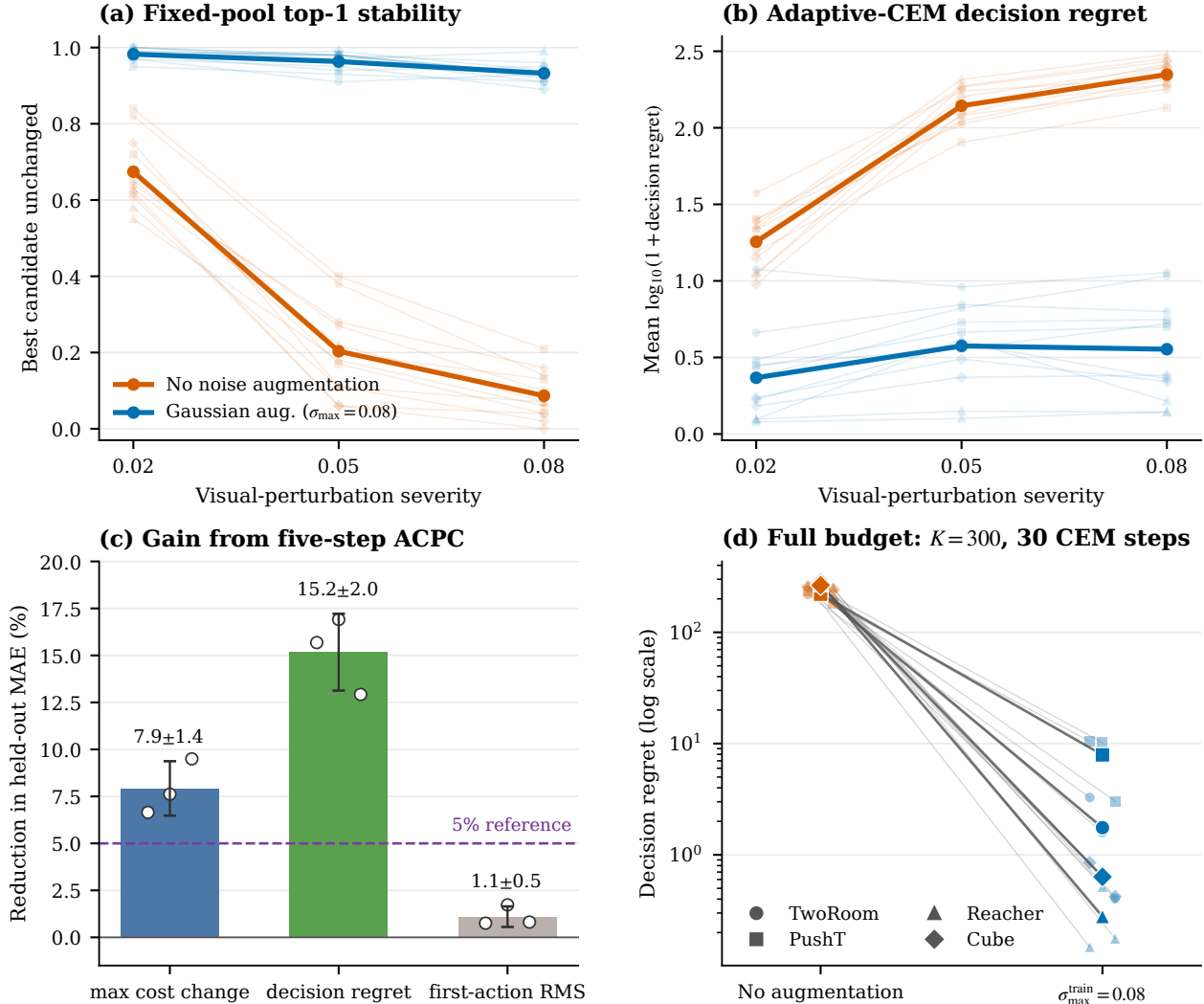


Figure 3: ACPC and CEM decision statistics. Panels (a)–(c) use 100 histories per task, run, and training condition: (a) best-candidate retention in a shared pool, (b) extra clean-history latent goal cost of the perturbed CEM decision, and (c) held-out MAE reduction from adding five-step ACPC. Panel (d) reports decision regret on 16 shared histories per task at the deployed $K = 300$, 30-step budget. Faint marks show task-run cells, heavy marks show equal-task summaries, and error bars are sample SD across runs. Decision regret is a model cost, not task success.

At perturbation severity 0.08, the task-level mean rate of retaining the best fixed-pool candidate is 0.92–0.96 after training with Gaussian augmentation, compared with 0.02–0.13 without augmentation. Mean positive decision regret is also lower with augmentation in every task-run cell: 3.64 ± 0.68 versus 227.04 ± 16.92 at reduced budget and 2.65 ± 1.43 versus 244.89 ± 27.08 at full budget (equal-task mean $\pm$ sample SD across runs). Because latent cost scales differ by task, these contrasts are reported descriptively rather than pooled as one effect. Appendix G gives the task-level values.

4.5 Cross-task threshold transfer

We select ATR and SMPR thresholds on every nonempty proper subset of the four tasks and evaluate them on all remaining tasks. This gives 14 directional partitions (Fig. 4).

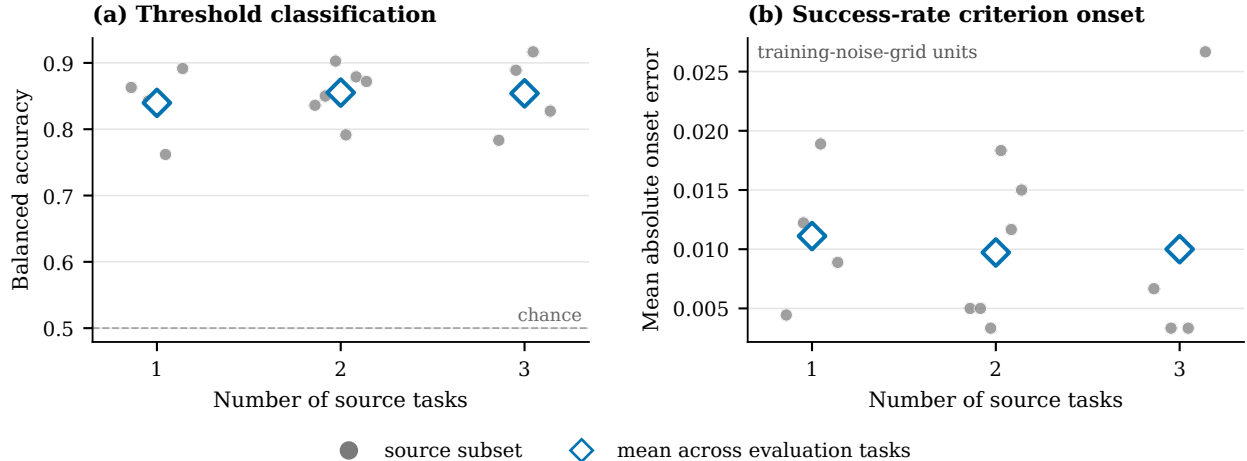


Figure 4: Cross-task evaluation over all 14 source/evaluation partitions. ATR and SMPR thresholds are selected using only the source tasks and applied unchanged to the remaining tasks. Small points show individual partitions and diamonds show the equal-task mean for each number of source tasks. Onset error is the absolute difference between the first augmentation level selected by the diagnostic and by the success-rate criterion.

Balanced accuracy is 0.840, 0.855, and 0.854 when thresholds are selected from one, two, and three source tasks, respectively. Corresponding precision/recall values are 0.903/0.824, 0.904/0.865, and 0.904/0.862. Mean absolute onset errors are 0.011, 0.010, and 0.010 Gaussian-augmentation-grid units, with maximum error 0.030 at each source coverage. Results do not improve monotonically as source tasks are added; which tasks provide calibration data matters, particularly when TwoRoom is held out. Thus the evaluation supports transfer of the threshold-selection procedure across these tasks, but not one numerical threshold for every task or model family. Appendix D reports every partition.

4.6 Application to PLDM

We apply the same paired-history construction, eight-step rollout, ATR normalization, SMPR guard, success-rate criterion, and threshold-selection procedure to the complete PLDM Gaussian sweep. The analysis uses PLDM’s encoder and predictor but does not depend on LeWM layers or training losses. For each PLDM evaluation task, thresholds are selected on the other three PLDM tasks and then fixed for its nine checkpoints.

Pooling the four evaluation branches gives 0.836 balanced accuracy, 0.789 precision, and 0.882 recall. Task balanced accuracies are 0.938, 0.750, 0.500, and 0.875 for TwoRoom, PushT, Reacher, and Cube, respectively; Reacher is therefore the unresolved boundary case. After normalizing ATR within each PLDM task, applying the corresponding LeWM source-task thresholds gives 0.807 balanced accuracy, 0.778 precision, and 0.824 recall. The shared equations and evaluation procedure therefore apply to both LeWM and PLDM, while the raw threshold values remain model-family specific.

Table 3: Applying the same ACPC analysis to the complete PLDM sweep of four tasks and nine augmentation levels. The first row selects thresholds on the other three PLDM tasks. The second uses the corresponding LeWM thresholds after normalizing ATR within each PLDM task. Metrics include all 36 PLDM evaluation rows; raw thresholds are not assumed to match across model families.

Calibration	BA	Precision	Recall	False pass	False miss
PLDM thresholds, other three tasks	0.836	0.789	0.882	4	2
LeWM thresholds, PLDM-normalized	0.807	0.778	0.824	4	3

4.7 Transfer to blur and resize

For each task, we apply the thresholds selected on the other three Gaussian-noise tasks to checkpoint pairs evaluated under blur and resize. No blur or resize result is used to select or adjust a threshold.

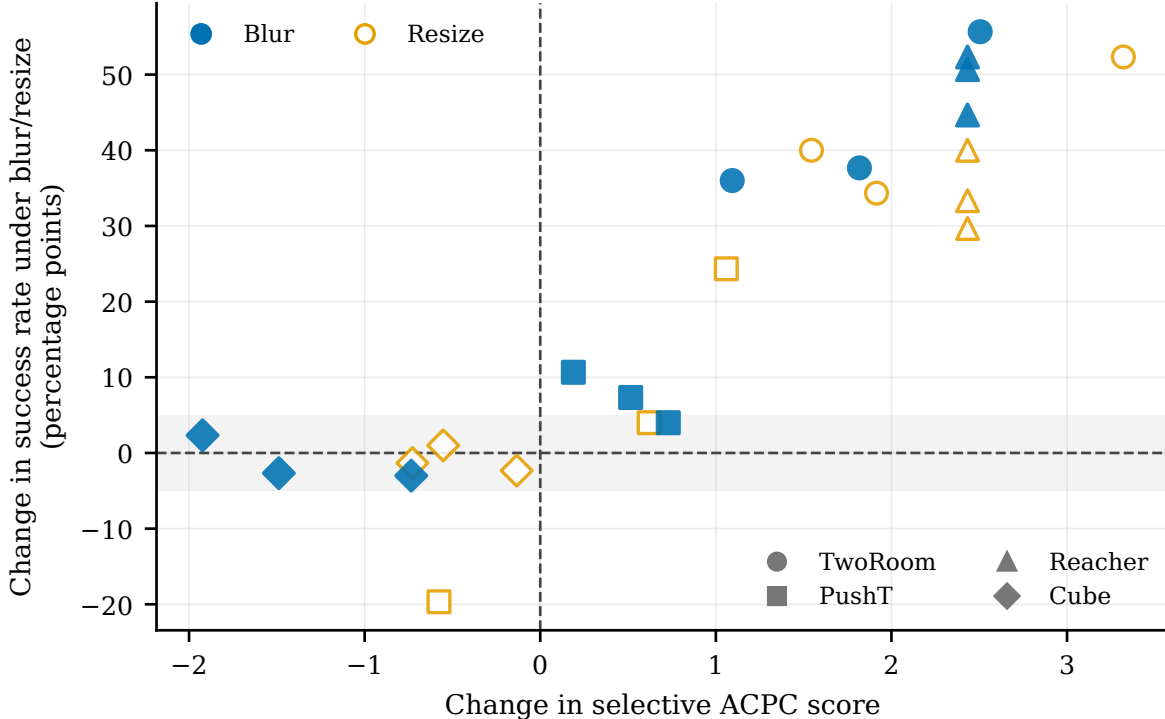


Figure 5: Selective-score and planning-success changes for all 24 LeWM checkpoint pairs under blur and resize. Each point compares a checkpoint trained with Gaussian augmentation ($\sigma_{\max} = 0.08$) with the corresponding checkpoint trained without augmentation. The horizontal axis uses thresholds selected on the other three Gaussian-noise tasks; the vertical axis reports the change in success rate under blur or resize. Marker shape denotes task and fill denotes the visual shift.

The sign rule reaches 0.889 balanced accuracy, 0.882 precision, and 1.000 recall, while the continuous score change has 0.909 Spearman correlation with the change in success rate. Blur and resize separately reach 0.875/0.900 balanced accuracy. The two discordant cases are both PushT changes of exactly four percentage points, which fall below the fixed five-point positive label while the selective score improves. These results support relative checkpoint ordering for these two visual shifts at the tested severity. They do not establish an absolute robustness classifier.

5 Discussion and limitations

The prediction experiment tests the quantity in Proposition 1 directly. Recorded-action eight-step ACPC predicts the change in latent error better than encoder shift, one-step ACPC, and same-horizon controls with zeroed or shuffled actions. Because every regression uses the same eight-step target, the result cannot be attributed only to comparing different rollout horizons. The direction is consistent across all tasks and runs, although the available runs do not support a precise estimate of population-level training variability.

The CEM experiment verifies the deterministic cost bound and tests whether candidate-conditioned ACPC explains decision changes beyond simpler features. Its incremental gain is consistent for decision regret, positive on three of four tasks for maximum cost change, and negligible for first-action RMS. These are model-cost and candidate-selection results, not environment success rates. The deployed-budget analysis also uses only 16 shared histories per task, and the evaluation does not include matched runs without diagnostic computation; it therefore does not estimate incremental wall-clock overhead.

Cross-task evaluation covers every source/remaining-task partition. The lack of monotonic improvement with source-set size shows that calibration depends on task composition. The PLDM experiment extends the equations and analysis procedure to a second joint-embedding dynamics architecture, while confirming that raw thresholds should remain model-family specific. Blur and resize test a narrower form of transfer: the Gaussian-calibrated score preserves useful relative checkpoint ordering at one severity for each visual shift.

Across the experiments, SMPR remains a gross-collapse guard rather than a semantic proof, the cost certificates require candidate-wise paired evaluation, and the adaptive result is conditional on pool alignment until an elite certificate fails. Stable but incorrect predictions remain possible, and SMPR cannot prove semantic validity. Important next steps are direct task outcomes for ACPC-informed planner interventions, multiple severities for blur and resize, richer different-state proxies, and additional model families with different objectives and planner interfaces.

6 Conclusion

ACPC measures how a same-state visual perturbation changes a world model’s predicted trajectory under fixed actions. Exact diagnostic inequalities relate this change to prediction error and squared goal cost, and a different-state margin guards against trivial collapse. Across four tasks, recorded-action ACPC adds information beyond one-step and same-horizon action controls, relates to CEM cost and decision changes, and supports thresholds that transfer to held-out tasks. The same analysis applies to PLDM, and a Gaussian-calibrated score retains useful checkpoint ordering under blur and resize. Numerical thresholds remain model-family specific.

References

- [1] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023.
- [2] Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhohus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.
- [3] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mido Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *Transactions on Machine Learning Research*, 2024.
- [4] Jiaheng Chen. ATM: Action-consistency transfer matrix for diagnosing and improving latent world models, 2026.
- [5] Yuxin Chen, Jianglan Wei, Chenfeng Xu, Boyi Li, Masayoshi Tomizuka, Andrea Bajcsy, and Ran Tian. Reimagination with test-time observation interventions: Distractor-robust world model predictions for visual model predictive control, 2025.
- [6] Tom Dupuis, Jaonary Rabarisoa, Quoc-Cuong Pham, and David Filliat. VIBR: Learning view-invariant value functions for robust visual control. In *Proceedings of The 2nd Conference on Lifelong Learning Agents*, volume 232 of *Proceedings of Machine Learning Research*, pages 658–682. PMLR, 2023.
- [7] T. W. Epps and Lawrence B. Pulley. A test for normality based on the empirical characteristic function. *Biometrika*, 70(3):723–726, 1983.
- [8] Carles Gelada, Saurabh Kumar, Jacob Buckman, Ofir Nachum, and Marc G. Bellemare. DeepMDP: Learning continuous latent space models for representation learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2170–2179. PMLR, 2019.
- [9] Hafez Ghaemi, Eilif Benjamin Muller, and Shahab Bakhtiari. seq-jepa: Autoregressive predictive learning of invariant-equivariant world models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- [10] Christopher Grimm, Andre Barreto, Satinder Singh, and David Silver. The value equivalence principle for model-based reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.
- [11] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640(8059):647–653, 2025.
- [12] Nicklas Hansen, Hao Su, and Xiaolong Wang. Td-mpc2: Scalable, robust world models for continuous control. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
- [13] Nicklas Hansen and Xiaolong Wang. Generalization in reinforcement learning by soft data augmentation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 13611–13617, 2021.
- [14] David Klindt, Yann LeCun, and Randall Balestriero. When does lejepa learn a world model?, 2026.
- [15] Ilya Kostrikov, Denis Yarats, and Rob Fergus. Image augmentation is all you need: Regularizing deep reinforcement learning from pixels. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- [16] Yann LeCun. A path towards autonomous machine intelligence. OpenReview preprint, 2022.
- [17] Etai Littwin, Omid Saremi, Madhu Advani, Vimal Thilak, Preetum Nakkiran, Chen Huang, and Joshua Susskind. How JEPA avoids noisy features: The implicit bias of deep linear self distillation networks, 2024.
- [18] Lucas Maes, Quentin Le Lidec, Luiz Facury, Nassim Massaudi, Ayush Chaurasia, Francesco Capuano, Richard Gao, Taj Gillin, Dan Haramati, Damien Scieur, Yann LeCun, and Randall Balestriero. stable-worldmodel: A platform for reproducible world modeling research and evaluation, 2026.
- [19] Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. Leworldmodel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026.
- [20] Mingyu Park, Samyeul Noh, Hyun Myung, and Donghwan Lee. Zero-shot visual generalization in model-based reinforcement learning via latent consistency. OpenReview (ICLR 2026 submission), 2025.
- [21] Bo-Kai Ruan, Teng-Fang Hsiao, Ling Lo, and Hong-Han Shuai. Is the future compatible? diagnosing dynamic consistency in world action models, 2026.
- [22] Finn Rasmus Schäfer, Korbinian Moller, Yuan Gao, Christian Oefinger, Sebastian Schmidt, and Johannes Betz. Imagined rollouts are kinematic, not dynamic: A diagnosis of long-horizon world-model failure, 2026. RSS Robot World Model Workshop 2026.
- [23] Gawon Seo, Dongwon Kim, and Suha Kwak. ACID: Action consistency via inverse dynamics for planning with world models, 2026.
- [24] Yutaka Shimizu and Masayoshi Tomizuka. Bisimulation metric for model predictive control. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- [25] Vlad Sobal, Jyothir S V, Siddhartha Jalagam, Nicolas Carion, Kyunghyun Cho, and Yann LeCun. Joint embedding predictive architectures focus on slow features, 2022.
- [26] Vlad Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Stress-testing offline reward-free reinforcement learning: A case for planning with latent dynamics models. In *WRL@ICLR 2025*, 2025.
- [27] Leonardo F. Toso, Davit Shadunts, Yunyang Lu, Nihal Sharma, Donglin Zhan, Nam H. Nguyen, and James Anderson. Learning invariant visual representations for planning with joint-embedding predictive world models. *arXiv preprint arXiv:2602.18639*, 2026.
- [28] Hugues Van Assel, Mark Ibrahim, Tommaso Biancalani, Aviv Regev, and Randall Balestriero. Joint embedding vs reconstruction: Provable benefits of latent space prediction for self-supervised learning, 2025.

- [29] Claas A. Voelcker, Anastasiia Pedan, Arash Ahmadian, Romina Abachi, Igor Gilitschenski, and Amir-Massoud Farahmand. Calibrated value-aware model learning with probabilistic environment models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 61745–61768. PMLR, 2025.
- [30] Zijie Wang, Wei Zhang, Weiming Zhang, Fanqi Zhang, Xiao Tan, Yipeng Qin, and Guanbin Li. World models as group actions, 2026.
- [31] Han Yan, Zishang Xiang, Zeyu Zhang, and Hao Tang. MWM: Mobile world models for action-conditioned consistent prediction, 2026.
- [32] Denis Yarats, Rob Fergus, Alessandro Lazaric, and Lerrel Pinto. Mastering visual continuous control: Improved data-augmented reinforcement learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.
- [33] Amy Zhang, Rowan T. McAllister, Roberto Calandra, Yarin Gal, and Sergey Levine. Learning invariant representations for reinforcement learning without reconstruction. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- [34] Zhenghao Zhang, Yuanxiang Wang, Zhenyu Guan, Yujia Yang, Bingkang Shi, Tianyu Zong, Hongzhu Yi, Guoqing Chao, Xingchen Chen, Tiankun Yang, Chenxi Bao, Tao Yu, Jingjing Zhou, and Jungang Xu. Delta-JEPA: Learning action-sensitive world models via latent difference decoding, 2026.

A Candidate-Cost and Fixed-Pool Analysis

Proof of Proposition 2. For one candidate, suppress the index and factor the squared-distance difference:

$$|\tilde{C} - C| = \left| \|\tilde{x} - g\|_2^2 - \|x - g\|_2^2 \right| \quad (12)$$

$$= \left| \langle \tilde{x} - x, \tilde{x} + x - 2g \rangle \right| \quad (13)$$

$$\leq \|\tilde{x} - x\|_2 \|\tilde{x} + x - 2g\|_2 \quad (14)$$

$$\leq r(\|x - g\|_2 + \|\tilde{x} - g\|_2) = b. \quad (15)$$

The first inequality is Cauchy–Schwarz and the second is the triangle inequality. Thus, for nominal winner w and competitor j ,

$$\tilde{C}_j - \tilde{C}_w \geq C_j - C_w - |\tilde{C}_j - C_j| - |\tilde{C}_w - C_w| \geq C_j - C_w - b_j - b_w.$$

For the perturbed winner $\tilde{w}$,

$$C_{\tilde{w}} \leq \tilde{C}_{\tilde{w}} + b_{\tilde{w}} \leq \tilde{C}_w + b_{\tilde{w}} \leq C_w + b_w + b_{\tilde{w}},$$

which proves Equation (4); nonnegativity follows from the optimality of w on the nominal pool. Positivity of Equation (5) makes every perturbed competitor strictly more costly than w . The same argument for every $i \in \mathcal{E}$ and $j \notin \mathcal{E}$ proves that Equation (6) preserves the exact elite set. Strict inequalities avoid relying on implementation-specific tie breaking.

Proof of Corollary 1. At iteration zero, the proposals are identical by assumption. Common random numbers therefore produce the same ordered action pool. If the elite certificate holds, both branches select the same action vectors as elites. The CEM mean/variance update is a deterministic function of those vectors, so the next proposals are identical. Repeating this argument establishes pool and proposal alignment by induction. If a certificate fails, equality of the elite sets is unknown; the induction stops and no later shared-pool statement is made.

Relation to the weighted rollout radius. The planner bound uses the displacement at the cost readout horizon. For the weighted stack in Equation (1), $\sqrt{\alpha_H} r_j \leq \text{ACPC}_H$ whenever $\alpha_H > 0$. The formal planner panel records r_j directly, avoiding the looser $1/\sqrt{\alpha_H}$ conversion. This does not require a future target, but it does require evaluating the candidate under both matched histories.

Table 4: Summary of the nine-level Gaussian-augmentation sweep. “Best” denotes the augmentation level with the highest mean planning success rate at evaluation noise $\sigma = 0.08$; Fig. 1 shows every level. Relative ATR is divided by its value without noise augmentation and is lower-is-better; SMPR is reported on its original scale and is higher-is-better.

Task	No-aug. success (%)	Best success (%)	Best $\sigma_{\max}$	Relative ATR	SMPR	Levels meeting criterion
TwoRoom	68.8	97.1	0.08	1.00→0.07	0.34→0.99	0.01–0.08
PushT	7.2	86.8	0.06	1.00→0.10	0.44→1.00	0.03–0.08
Reacher	18.2	83.3	0.07	1.00→0.03	0.73→0.99	0.03–0.08
Cube	43.1	66.0	0.03	1.00→0.31	0.45→0.99	0.03–0.07

Sharpness without additional assumptions. Neither principal bound admits a uniformly smaller right-hand side from the available quantities alone. For a unit vector u , common target $Y = 0$, and two stacked predictions au and bu with $a, b \geq 0$, the reverse-triangle bound in Equation (2) holds with equality. Likewise, setting $g = 0$, $x = au$, and $\tilde{x} = bu$ makes $|\tilde{C} - C| = |b - a|(a + b) = r(\|x - g\|_2 + \|\tilde{x} - g\|_2)$.

B Evaluation and Diagnostic Protocol

These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube with evaluation seeds 42/43/44 and 100 episodes per seed. The main evaluation perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Checkpoints trained with noise use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \dots, 0.08\}$.

PLDM uses one independently trained model family and the same four-task, nine-checkpoint Gaussian grid, evaluation seeds, trajectory count, paired history construction, $H = 8$ rollout statistic, and SMPR definition. For its cross-task analysis, ATR is divided by its value for the corresponding PLDM checkpoint trained without augmentation, and thresholds are selected on the other three PLDM tasks. The evaluation task’s success rates never enter threshold selection. A second test applies the corresponding LeWM threshold pair after this within-task PLDM normalization. Raw thresholds are not assumed to be shared across model families.

Within each task and training run, let B and M be the success rates at evaluation noise $\sigma = 0.08$ for the checkpoint trained without augmentation and the best checkpoint in the sweep. A checkpoint meets the success-rate criterion when its noisy-observation success rate is at least $B + 0.8(M - B)$ and its clean success rate is no more than five percentage points below that of the checkpoint trained without augmentation. The green spans in Figure 1 require this criterion for a majority of runs.

ATR is computed for each task, training run, and checkpoint from one weighted-stacked horizon radius per anchor. The default is $H = 8$ with uniform $\alpha_k = 1/H$. Each radius is divided by that anchor’s q50 clean observed-transition scale, including the transition from the last history observation to the first future observation; multiple noise draws are first averaged within the anchor, and q90 is then taken across anchors. For the sweep display and within-family cross-task calibration, this checkpoint statistic is also divided by the corresponding value without noise augmentation before thresholds or run summaries are computed. SMPR uses programmatic label-crossing state pairs, the q90 same-state radius, the closest-35% neighborhood, and normalized margin 0.10. Across-run means and dispersions are descriptive rather than population inference.

For cross-task evaluation, candidate thresholds are $t_R \in \{0.05, 0.075, 0.10, 0.15, 0.20, 0.30\}$ for task-relative ATR and $t_M \in \{0.80, 0.85, 0.90, 0.95\}$ for SMPR. Within each source-task subset, the selection criterion lexicographically minimizes mean absolute onset error, false-early and false-late counts, then maximizes balanced accuracy. The selected pair is applied without modification to every evaluation task. No evaluation-task label or blur/resize outcome enters threshold selection.

All summary figures and tables are rebuilt deterministically by the scripts documented in `paper1/scripts/REA DME.md`. Checkpoint-level diagnostics additionally require the released checkpoints and datasets. Machine-readable summaries record training seeds, evaluation seeds, and protocol hashes.

C Prediction-Error Scope and Controls

The common-future inequality in Proposition 1 has zero numerical violations in every logged eight-step and candidate-conditioned five-step task×run×training-condition cell. This is an implementation invariant, not an independent statistical success count.

Table 5: Predicting the change in eight-step latent error under a visual perturbation. All regressions use the same target and base features; they differ only in the ACPC feature. The recorded-action eight-step feature is compared with one-step ACPC and with the strongest same-horizon feature using zeroed or shuffled actions. Values are relative reductions in held-out MAE after rotating evaluation over 16 trajectory groups; higher is better.

Training run	vs. one-step ACPC	vs. same-horizon control	Task-run cells improved
seed 3072	55.5%	51.4%	4/4
seed 3073	60.8%	54.8%	4/4
seed 3074	51.4%	47.7%	4/4
mean $\pm$ SD	55.9 $\pm$ 4.7%	51.3 $\pm$ 3.5%	12/12

Table 6: Held-out MAE for predicting the absolute change in eight-step latent prediction error. Each row contains 16 trajectory groups. The recorded-action column uses eight-step ACPC with the observed actions; the control column uses the strongest eight-step feature with zeroed or shuffled actions. Lower is better.

Task	seed	one-step	same-horizon control	recorded-action 8-step	control type	win blocks
TwoRoom	3072	2.535	2.099	0.755	time-shuffled actions	15/16
TwoRoom	3073	2.336	2.015	0.866	time-shuffled actions	15/16
TwoRoom	3074	2.075	1.911	1.006	cross-trajectory actions	13/16
PushT	3072	2.068	2.068	1.762	time-shuffled actions	11/16
PushT	3073	1.969	2.008	1.315	zero actions	13/16
PushT	3074	1.909	1.833	1.439	zero actions	13/16
Reacher	3072	1.358	1.097	0.288	time-shuffled actions	16/16
Reacher	3073	1.399	0.793	0.247	time-shuffled actions	16/16
Reacher	3074	1.409	1.136	0.263	time-shuffled actions	16/16
Cube	3072	1.171	1.037	0.489	zero actions	14/16
Cube	3073	1.416	1.212	0.500	zero actions	14/16
Cube	3074	1.061	1.008	0.552	time-shuffled actions	14/16

Logged-future prediction and candidate planning are different estimands: the former predicts an independent action-matched observed future after common covariates are supplied, whereas the latter compares candidates from the actual planner pool. We do not infer one result from the other. The separate panel in Section 4.4 supplies planner-facing evidence with same-pool, candidate-conditioned one- and five-step features.

The prediction-error analysis is restricted to checkpoints trained without noise augmentation and to the listed visual probes. Every run improves on all four tasks; the across-run mean and sample standard deviation summarize repeatability rather than a precisely estimated population distribution.

D Cross-Task Threshold Partitions

The 14 rows below are exhaustive: selecting one, two, or three of four tasks as sources creates $4 + 6 + 4$ directional partitions. Each evaluation task retains every training run and all nine checkpoints. The apparent sample size is therefore not inflated by treating checkpoint rows as independent observations.

E Cross-Stressor Pairs and Discordant Cases

Both discordant rows lie near the success-rate criterion: PushT run 3072 under resize and PushT run 3074 under blur each improve success rate by four percentage points, one point below the five-point criterion, while their selective scores increase. The complete table shows these cases rather than only the aggregate classification metrics.

Table 7: Cross-task evaluation of the selective ACPC rule. Thresholds are selected on the source tasks and applied unchanged to all remaining tasks. Metrics first average training runs within each evaluation task and then weight tasks equally. Onset error is measured in Gaussian-augmentation grid units.

Source tasks	Evaluation tasks	τ_R	τ_M	BA	P / R	Onset error mean / max
TwoRoom	PushT, Reacher, Cube	0.3	0.90	0.863	0.872 / 0.931	0.004 / 0.010
PushT	TwoRoom, Reacher, Cube	0.2	0.95	0.842	0.906 / 0.816	0.012 / 0.030
Reacher	TwoRoom, PushT, Cube	0.15	0.95	0.762	0.883 / 0.655	0.019 / 0.030
Cube	TwoRoom, PushT, Reacher	0.3	0.95	0.892	0.950 / 0.894	0.009 / 0.020
TwoRoom, PushT	Reacher, Cube	0.3	0.90	0.836	0.836 / 0.925	0.005 / 0.010
TwoRoom, Reacher	PushT, Cube	0.3	0.90	0.850	0.856 / 0.897	0.005 / 0.010
TwoRoom, Cube	PushT, Reacher	0.3	0.90	0.903	0.925 / 0.972	0.003 / 0.010
PushT, Reacher	TwoRoom, Cube	0.2	0.95	0.791	0.883 / 0.725	0.018 / 0.030
PushT, Cube	TwoRoom, Reacher	0.3	0.95	0.879	0.952 / 0.869	0.012 / 0.020
Reacher, Cube	TwoRoom, PushT	0.2	0.95	0.872	0.972 / 0.800	0.015 / 0.030
TwoRoom, PushT, Reacher	Cube	0.3	0.90	0.783	0.767 / 0.850	0.007 / 0.010
TwoRoom, PushT, Cube	Reacher	0.3	0.90	0.889	0.905 / 1.000	0.003 / 0.010
TwoRoom, Reacher, Cube	PushT	0.3	0.90	0.917	0.944 / 0.944	0.003 / 0.010
PushT, Reacher, Cube	TwoRoom	0.2	0.95	0.827	1.000 / 0.655	0.027 / 0.030

Table 8: Transfer of the selective ACPC score from Gaussian noise to blur and resize. For each task, thresholds are selected on the other three Gaussian-noise tasks and then fixed. The table compares the score change with the change in planning success rate between checkpoints trained with and without Gaussian augmentation.

Visual shift	Pairs	BA	Precision / recall	Spearman	Discordant
All pairs	24	0.889	0.882 / 1.000	0.909	2
Blur	12	0.875	0.889 / 1.000	0.937	1
Resize	12	0.900	0.875 / 1.000	0.859	1

Table 9: All 24 LeWM checkpoint pairs evaluated under blur and resize. ATR_{rel} and SMPR are measured for the checkpoint trained with Gaussian augmentation. ΔP is its change in planning success rate relative to the checkpoint trained without augmentation, and ΔS is the corresponding selective-score change. A success-rate change is positive when $\Delta P \geq 5$ percentage points and the clean success rate decreases by at most five points. Outcome lists the success-rate classification followed by the score classification.

Task	Seed	Shift	ATR_{rel}	SMPR	ΔP	ΔS	Outcome
TwoRoom	3072	blur	0.499	0.885	55.7	2.505	positive / positive
TwoRoom	3072	resize	0.336	0.967	52.3	3.321	positive / positive
TwoRoom	3073	blur	0.781	0.262	36.0	1.093	positive / positive
TwoRoom	3073	resize	0.691	0.361	40.0	1.544	positive / positive
TwoRoom	3074	blur	0.636	0.541	37.7	1.818	positive / positive
TwoRoom	3074	resize	0.617	0.656	34.3	1.915	positive / positive
PushT	3072	blur	0.943	0.939	10.7	0.189	positive / positive
PushT	3072	resize	0.814	0.969	4.0	0.620	nonpositive / positive
PushT	3073	blur	0.846	0.918	7.3	0.515	positive / positive
PushT	3073	resize	0.682	0.969	24.3	1.060	positive / positive
PushT	3074	blur	0.781	0.959	4.0	0.729	nonpositive / positive
PushT	3074	resize	1.173	0.939	-19.7	-0.577	nonpositive / nonpositive
Reacher	3072	blur	0.155	0.990	50.7	2.433	positive / positive
Reacher	3072	resize	0.210	0.990	29.7	2.433	positive / positive
Reacher	3073	blur	0.176	0.990	52.3	2.433	positive / positive
Reacher	3073	resize	0.207	0.990	40.0	2.433	positive / positive
Reacher	3074	blur	0.190	0.990	44.7	2.433	positive / positive
Reacher	3074	resize	0.195	0.990	33.3	2.433	positive / positive
Cube	3072	blur	1.447	0.330	-2.7	-1.488	nonpositive / nonpositive
Cube	3072	resize	1.166	0.530	1.0	-0.552	nonpositive / nonpositive
Cube	3073	blur	1.577	0.260	2.3	-1.923	nonpositive / nonpositive
Cube	3073	resize	1.218	0.560	-1.3	-0.728	nonpositive / nonpositive
Cube	3074	blur	1.220	0.660	-3.0	-0.735	nonpositive / nonpositive
Cube	3074	resize	1.040	0.770	-2.3	-0.134	nonpositive / nonpositive

F Local Gaussian Sensitivity as a Mechanistic Interpretation

For a small isotropic observation perturbation $\delta x \sim \mathcal{N}(0, \sigma^2 I)$, first-order linearization of encoder E followed by rollout map G gives

$$\mathbb{E}[\|\Delta G\|_2^2] \approx \sigma^2 \|J_G J_E\|_F^2. \quad (16)$$

This explains one mechanism by which Gaussian noise training can contract ACPC: it reduces the composed local sensitivity of the encoder–predictor path. It is a local approximation, not a global robustness bound and not a separate contribution.

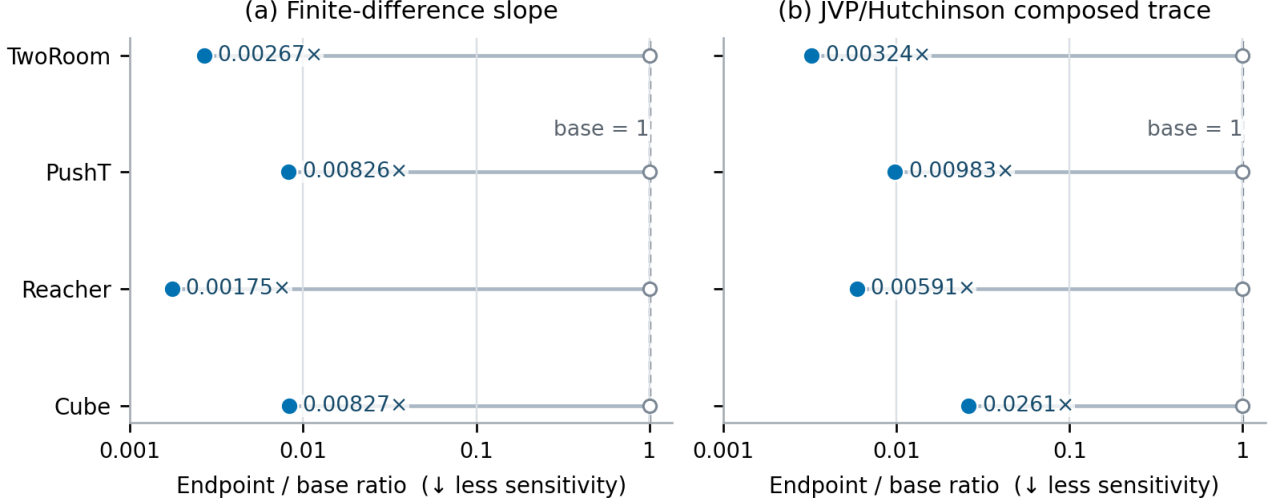


Figure 6: Local-sensitivity ratio of the checkpoint trained with Gaussian augmentation to the checkpoint trained without it, estimated by finite differences and by a JVP/Hutchinson estimate of the composed Jacobian trace. Values below one indicate contraction after augmentation. The two estimators agree in direction for every task; these ratios do not establish task performance.

G Fixed-Pool Candidate Stability

The CEM analysis does not use task-success labels. It compares checkpoints trained without augmentation and with $\sigma_{\max} = 0.08$ on all four tasks. Fixed-pool shards capture the ordered initial proposal, evaluate the nominal and perturbed histories through one shared cost path, and record exact same-winner and elite-set events for 100 histories per task, run, and training condition at severities 0, 0.02, 0.05, 0.08. Adaptive shards use a common initial proposal and common random numbers. Proposal alignment is asserted only while all preceding elite updates remain certified.

For nominal final plan $\mathbf{a}$ and perturbed plan $\tilde{\mathbf{a}}$, the reported positive clean-history regret is

$$[C_h(\tilde{\mathbf{a}}) - C_h(\mathbf{a})]_+.$$

It measures decision degradation in the model’s latent goal cost, not an environment outcome. First-action RMS is reported separately. The full-budget evaluation keeps the same definitions but uses 16 fixed histories per task shared across training conditions and runs, $K = 300$, top-30 elites, and 30 CEM steps. Whole-shard runtimes and forward-call counts are retained for provenance, but without matched timing runs that omit the diagnostic they do not estimate incremental overhead.

Across 9,600 joined reduced-budget rows there are zero squared-distance-bound violations, zero false top-1 certificates, and zero false elite certificates. Identity probes have zero five-step displacement and zero first-action RMS; all identity updates remain aligned. These checks establish implementation soundness and are not additional independent statistical samples.

At severity 0.08, the sufficient top-1 certificate covers 10–70% of histories from checkpoints trained with augmentation across task–run blocks (and none from checkpoints trained without augmentation); elite-set coverage is 0–30% with augmentation. Coverage is lower than realized stability because these conditions are sufficient but not necessary.

Table 10: CEM decision statistics at Gaussian perturbation severity 0.08. Values are mean $\pm$ sample SD across independent training runs after histories are averaged within each run. Reduced-budget rows use 100 histories, $K = 64$, top-8 elites, and eight CEM steps; full-budget rows use the same 16 histories per task with $K = 300$, top-30 elites, and 30 steps. Regret is measured with the model’s squared latent goal cost and is not an environment success rate.

Task	Best candidate unchanged		Reduced-budget regret		Full-budget regret	
	No aug.	$\sigma_{\max}=0.08$	No aug.	$\sigma_{\max}=0.08$	No aug.	$\sigma_{\max}=0.08$
TwoRoom	0.13 $\pm$ 0.03	0.93 $\pm$ 0.02	203.81 $\pm$ 9.97	8.47 $\pm$ 2.75	224.97 $\pm$ 30.56	1.75 $\pm$ 1.44
PushT	0.13 $\pm$ 0.09	0.92 $\pm$ 0.01	167.50 $\pm$ 29.53	4.30 $\pm$ 0.28	220.14 $\pm$ 32.97	7.91 $\pm$ 4.23
Reacher	0.07 $\pm$ 0.01	0.96 $\pm$ 0.04	281.12 $\pm$ 20.24	0.47 $\pm$ 0.14	269.44 $\pm$ 22.21	0.28 $\pm$ 0.20
Cube	0.02 $\pm$ 0.02	0.92 $\pm$ 0.03	255.72 $\pm$ 12.28	1.31 $\pm$ 0.11	265.03 $\pm$ 28.08	0.63 $\pm$ 0.22
Equal-task mean	0.09 $\pm$ 0.03	0.93 $\pm$ 0.01	227.04 $\pm$ 16.92	3.64 $\pm$ 0.68	244.89 $\pm$ 27.08	2.65 $\pm$ 1.43

H Qualitative Neighborhood Visualization

The two-dimensional view complements the quantitative analysis by showing how repeated perturbations of the same anchors are arranged before and after noise training. Because t-SNE does not preserve metric geometry, all reported evidence remains in the original projected-rollout space.

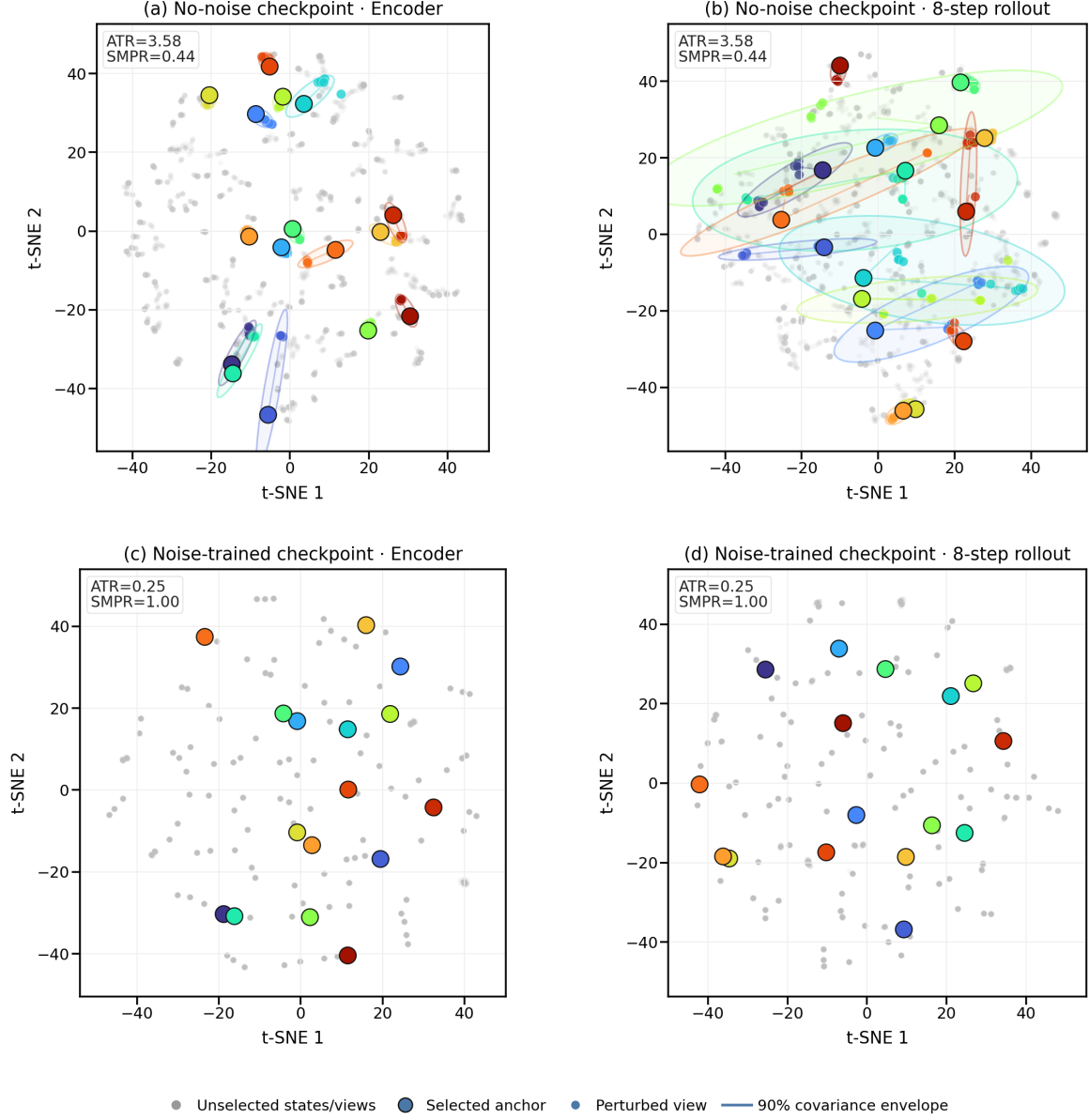


Figure 7: Qualitative PushT neighborhoods for matched perturbations at checkpoints trained without augmentation and with $\sigma_{\max} = 0.08$, shown in encoder and eight-step rollout spaces. t-SNE is not metric preserving: no displayed distance, area, or overlap enters a claim; ATR and SMPR are computed in the original space.